

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456967.7824 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.048
Transit depth (Rp/Rs)^2	1.25 +/- 1.08 [%]
Orbital Inclination (inc)	85.58 +/- 2.7
Airmass coefficient 1 (a1)	0.946 +/- 0.022
Airmass coefficient 2 (a2)	0.0298 +/- 0.0061
Scatter in the residuals of the lightcurve fit is	2.34 %
Best Comparison Star	#2 - [593, 301]
Optimal Aperture	2.27
Optimal Annulus	14.39
Transit Duration (day)	0.065 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.112 ± 0.048 vs 0.1646 (deviation 1.1σ)
Duration fit vs published	0.065 d vs 0.0758 d (deviation 0.52σ)
Mid-time O–C	-3.7 min
Depth SNR	2.3
Residual scatter	2.34 %

Light curve

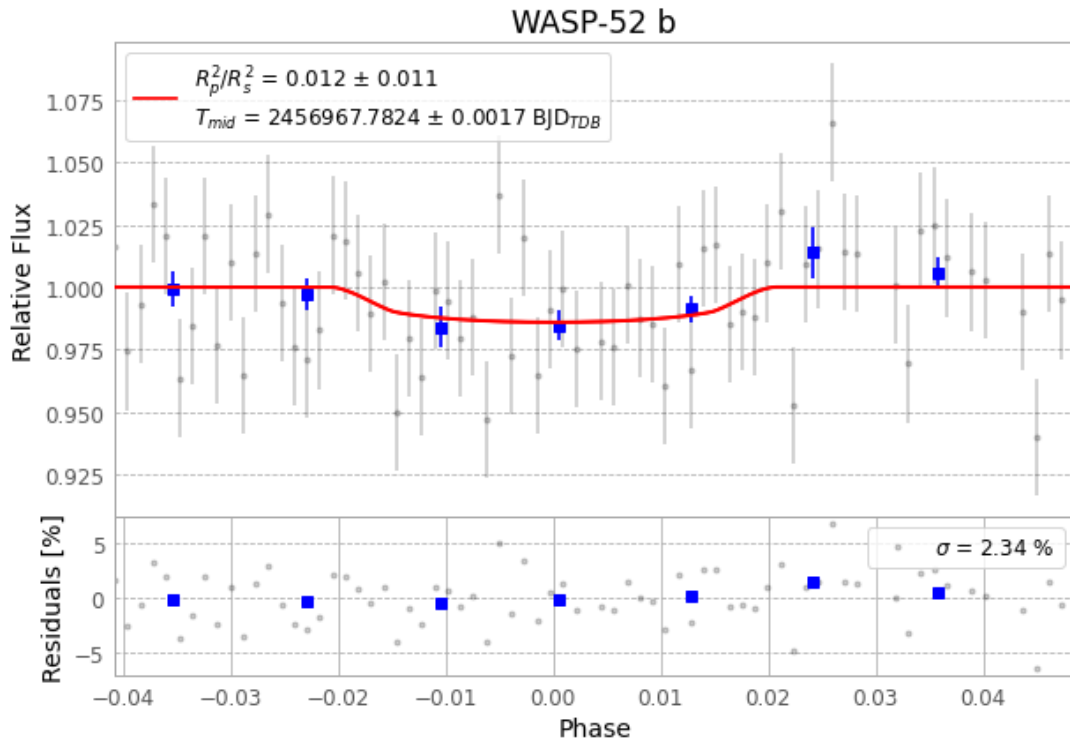


Figure 1 — FinalLightCurve_WASP-52 b_2014-11-05.png (SHA-256 1f0e2bd50856c076...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [516, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a3c10d5759437733...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-52141106045712.FITS → WASP-52141106084212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-141106.log (961008434ea5...), FinalLightCurve_WASP-52 b_2014-11-05.png (1f0e2bd50856...), FinalLightCurve_WASP-52 b_2014-11-05.csv (3f594c66d2ff...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 21:33Z.